Dingling Yao

Final-year ELLIS PhD Candidate in Computer Science, specializing in **interpretable representation learning for scientific discovery**. Experience with scalable Python/JAX/PyTorch workflows; published at NeurIPS, ICML, and ICLR.

Email: yaodingling806@gmail.com **Website**: ddcoan.github.io **GitHub**: github.com/DDCoan
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Education

ELLIS PhD in Computer Science **Apr 2022 – Present**
Institute of Science and Technology Austria (ISTA) and Max Planck Institute for Intelligent Systems (MPI-IS) *Austria & Germany*
Topic: Interpretable Causal Representation Learning for Scientific Discovery.
PhD Advisor: Prof. Francesco Locatello **Co-advisor**: Prof. Georg Martius

M.Sc. in Machine Learning **Oct 2020 – Mar 2022**
University of Tübingen *Tübingen, Germany*
Grade: *Sehr gut (very good)*
Relevant courses: Probabilistic Machine Learning, Deep Learning, Time Series and Dynamical Systems.

Industrial Research Experience

*Experience spanning **fundamental research** (Amazon) and **applied research** (BCAI), with joint publications at top-tier venues.*

Amazon Web Services **Oct 2022 – Mar 2023**
Research Intern *Tübingen, Germany*
Investigated **causal learning algorithms** for intrinsic robot exploration, combining representation learning and control.
Implemented scalable deep learning architectures in PyTorch for high-dimensional sensorimotor data, improving sample efficiency.

Bosch Center for Artificial Intelligence **May 2020 – Jun 2021**
Part-time ML Researcher *Renningen, Germany*
Developed **probabilistic models** for experimental design and adaptive data collection in industrial settings.

Awards & Fellowships

Outstanding academic contributions, as shown by publications at top-tier conferences and industrial fellowships.

Google PhD Fellowship **2025-2026**
Machine Learning and ML Foundations

Amazon-MPI Fellowship **2022-2023**
Amazon-MPI Science Hub

Leadership

NeurIPS Workshop Organization **2025**
Lead Organizer *CauScien Workshop at NeurIPS 2025*
Successfully organized the **first** CauScien Workshop at NeurIPS 2025 with more than 300 participants.

Speaker Series **2025**
Organizer *ELLIS x UniReps*
Co-organizing the ELLIS x UniReps Speaker Series on unifying representations with **over 100** participants.

Technical Skills

Climate & Scientific ML: Bulk ERA5/CMIP retrieval & preprocessing (CDS API), lightweight atmospheric modeling (SpeedyWeather.jl), scalable workflows with xarray/Zarr/WebDataset for multi-TB datasets

Languages & Frameworks: Python, PyTorch, JAX, scikit-learn, Julia, SQL

ML Expertise: Interpretable Representation Learning, Causal Inference, Generative Models, Reinforcement Learning (basic)

Cloud & Infrastructure: AWS, S3, HPC clusters

Invited Talks

Dedicated to impactful international research communication, evidenced by worldwide invited talks.

Causal Representation Learning for Scientific Discovery **Apr 2025**
Institute of Computing Technology, Chinese Academy of Sciences *Remote*

Multi-view Causal Representation Learning with Partial Observability **Mar 2024**
SIAM Conference on Uncertainty Quantification (UQ24) *Trieste, Italy*

Invited Talk **Feb 2024**
Bellairs Workshop on Causality *Barbados*

Oral Presentation **Dec 2023**
Causal Representation Learning workshop at NeurIPS 2023 *New Orleans, US*

Publications and Preprints

My research focuses on developing scalable methods for interpretable representation learning from high-dimensional, real-world scientific datasets. By learning compact and semantically meaningful representations, my work enhances model interpretability and enables robust downstream scientific analyses across diverse domains, including climate science [6] and ecological studies [2].

Publications

[1] J. Chen, **D. Yao**, A. Pervez, D. Alistarh, and F. Locatello. "Scalable Mechanistic Neural Networks". In: *International Conference on Learning Representations* ***(ICLR 2025)***. 2025. URL: https://arxiv.org/abs/2410.06074.

[2] **D. Yao**, D. Rancati, R. Cadei, M. Fumero, and F. Locatello. "Unifying Causal Representation Learning with the Invariance Principle". In: *International Conference on Learning Representations* ***(ICLR 2025)***. 2025. URL: https://arxiv.org/abs/2409.02772.

[3] **D. Yao**, F. Tronarp, and N. Bosch. "Propagating Model Uncertainty through Filtering-based Probabilistic Numerical ODE Solvers". In: *International Conference on Probabilistic Numerics 2025* ***(ProbNum 2025, oral)***. 2025. URL: https://www.arxiv.org/abs/2503.04684.

[4] **D. Yao***, S. Huang*, R. Cadei, K. Zhang, and F. Locatello. "The Third Pillar of Causal Analysis? A Measurement Perspective on Causal Representations". In: *Advances in Neural Information Processing Systems* ***(NeurIPS 2025)***. 2025. URL: https://arxiv.org/abs/2505.17708.

[5] D. Xu, **D. Yao**, S. Lachapelle, P. Taslakian, J. von Kügelgen, F. Locatello, and S. Magliacane. "A sparsity principle for partially observable causal representation learning". In: *International Conference on Machine Learning* ***(ICML 2024)***. 2024. URL: https://openreview.net/pdf?id=MKGrRVODWR.

[6] **D. Yao**, C. Muller, and F. Locatello. "Marrying Causal Representation Learning with Dynamical Systems for Science". In: *Advances in Neural Information Processing Systems* ***(NeurIPS 2024)***. 2024. URL: https://arxiv.org/abs/2405.13888.

[7] **D. Yao**, D. Xu, S. Lachapelle, S. Magliacane, P. Taslakian, G. Martius, J. von Kügelgen, and F. Locatello. "Multi-view causal representation learning with partial observability". In: *International Conference on Learning Representations* ***(ICLR 2024, spotlight, top 5%)***. 2024. URL: https://openreview.net/forum?id=OGtnhKQJms.

[8] H. S. A. Yu, **D. Yao**, C. Zimmer, M. Toussaint, and D. Nguyen-Tuong. "Active learning in Gaussian process state space model". In: *Machine Learning and Knowledge Discovery in Databases. Research Track: European Conference,* ***ECML PKDD 2021****, Bilbao, Spain, September 13–17, 2021, Proceedings, Part III 21*. Springer. 2021, pp. 346–361. URL: https://arxiv.org/abs/2108.00819.

*Equal contribution.